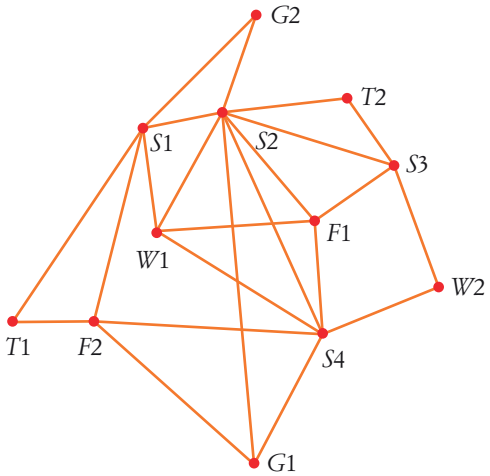
YEAR 12 GENERAL MATHEMATICS INVESTIGATIVE TASK 2 SOLUTIONS

Networks

Question 1

Consider this graph showing different locations at a music festival.



Key:

G: Gate

S: Stage

W: Water

F: Food

T: Toilet

a Identify the number of edges and vertices in this graph.

12 vertices, 23 edges

1 mark

b Which food stand(s) are adjacent to stage 2?

Food stand 1

1 mark

c Which stage is directly connected to all others?

Stage 2

1 mark

© Cengage Learning Australia Pty Ltd 2014

d What is the degree sum of this graph?

Degree sum $= 2 \times$ edges

$= 2 \times 23$

$= 46$

1 mark

e State which vertex has the highest degree. State the magnitude of this degree.

Stage 2, degree of 8

1 mark

f Create an adjacency matrix for the graph.

	$G1$	$G2$	$S1$	$S2$	$S3$	$S4$	$W1$	$W2$	$F1$	$F2$	$T1$	$T2$
$G1$	0	0	0	1	0	1	0	0	0	1	0	0
$G2$	0	0	1	1	0	0	0	0	0	0	0	0
$S1$	0	1	0	1	0	0	1	0	0	1	1	0
$S2$	1	1	1	0	1	1	1	0	1	0	0	1
$S3$	0	0	0	1	0	0	0	1	1	0	0	1
$S4$	1	0	0	1	0	0	1	1	1	1	0	0
$W1$	0	0	1	1	0	1	0	0	1	0	0	0
$W2$	0	0	0	0	1	1	0	0	0	0	0	0
$F1$	0	0	0	1	1	1	1	0	0	0	0	0
$F2$	1	0	1	0	0	1	0	0	0	0	1	0
$T1$	0	0	1	0	0	0	0	0	0	1	0	0
$T2$	0	0	0	1	1	0	0	0	0	0	0	0

1 mark

g Is there a stage that is not directly connected to food, water or a toilet? If so, state which stage, and to what it is not connected.

Stage 4 is not connected to a toilet.

2 marks

h Lisa enters gate 2, watches bands at stages 1 and 4 and then leaves. Define a walk that she could complete.

$G2-S1-W1-S4-G1$

$G2-S1-S2-S4-S2-G2$

2 marks

i Decide whether the graph is Eulerian or semi-Eulerian and hence define a closed or open trail that exists.

Graph has two vertices of an odd degree and all others are even, hence it is a semi-Eulerian graph and an open trail exists.

© Cengage Learning Australia Pty Ltd 2014

Trail must start at one of the vertices with an odd degree: $G1$ or $S1$.

One example is:

$S1-G2-S2-S1-W1-S2-T2-S3-S2-S4-W1-F1-S3-W2-S4-F1-S2-G1-S4-F2-S1-T1-F2-G1$

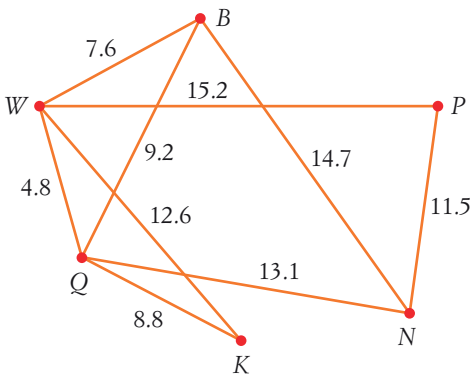
4 marks

Question 2

Consider this information showing the distances in kilometres between camping sites at a national park.

	Bandicoot Bend	Possum Place	Numbat Nest	Kangaroo Kips	Quiet Quoll	Wombat West
Bandicoot Bend	–	–	14.7	–	9.2	7.6
Possum Place	–	–	11.5	–	–	15.2
Numbat Nest	14.7	11.5	–	–	13.1	–
Kangaroo Kips	–	–	–	–	8.8	12.6
Quiet Quoll	9.2	–	13.1	8.8	–	4.8
Wombat West	7.6	15.2	–	12.6	4.8	–

a Draw a graph to represent the information in the table.



2 marks

b Which campsite is not connected to Wombat West?

Numbat Nest

1 mark

c Charlie hiked 11.2 km to Quiet Quoll and, over the next few days, hiked to Wombat West, Possum Place, Numbat Nest, back to Quiet Quoll and then out again. Assuming he followed the direct trails between each campsite, what distance did Charlie cover?

67 km

1 mark

d What is the shortest distance from Numbat Nest to Wombat West?

17.9 km, via Quiet Quoll

1 mark

© Cengage Learning Australia Pty Ltd 2014

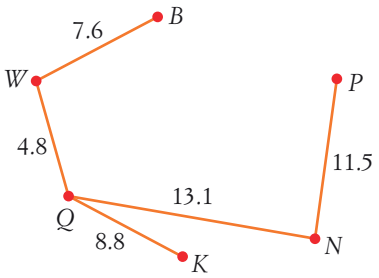
e Define the Hamiltonian cycle that exists and state the distance that it covers.

$Q-K-W-P-N-B-Q$

It covers $8.8 + 12.6 + 15.2 + 11.5 + 14.7 + 9.2 = 72$ km.

3 marks

f Use Prim's algorithm to find the minimum spanning tree and hence find the minimum distance between all campsites.

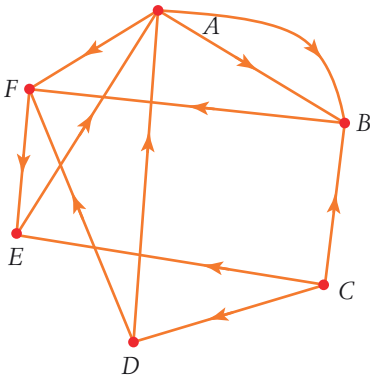


Minimum distance $= 7.6 + 4.8 + 8.8 + 13.1 + 11.5 = 45.8$ km.

3 marks

Question 3

Consider this directed network.



a Which locations are not accessible from *E*?

C and *D*

1 mark

b List all possible paths from *E*.

A, *AB*, *ABF*, *ABFE*, *AF*, *AFE*

2 marks

© Cengage Learning Australia Pty Ltd 2014

c Represent the one- and two-stage pathways in the directed network in matrix form.

$\mathbf{R}^1 =$

	A	B	C	D	E	F
A	0	2	0	0	0	1
B	0	0	0	0	0	1
C	0	1	0	1	1	0
D	1	0	0	0	0	1
E	1	0	0	0	0	0
F	1	0	0	0	0	0

$\mathbf{R}^2 =$

	A	B	C	D	E	F
A	0	0	0	0	1	1
B	0	0	0	0	1	0
C	2	0	0	0	0	2
D	0	2	0	0	1	1
E	0	2	0	0	0	1
F	1	0	0	0	0	0

2 marks

This graph actually represents teams in a hockey tournament.

d Who plays the most games?

A

1 mark

e Use your answer to part **c** to show who is the dominant team.

$D = \mathbf{R}^1 + \mathbf{R}^2 =$

	A	B	C	D	E	F
A	0	2	0	0	1	2
B	0	0	0	0	1	1
C	2	1	0	1	1	2
D	1	2	0	0	1	2
E	1	2	0	0	0	1
F	2	0	0	0	0	0

Team	Dominance
A	5
B	2
C	7
D	6
E	4
F	2

Team C is the dominant team.

2 marks

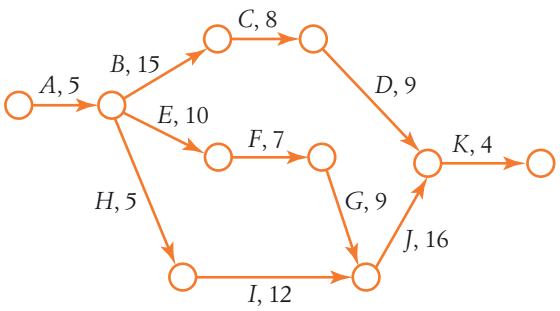
© Cengage Learning Australia Pty Ltd 2014

Question 4

Consider the information in this table.

Activity	Time (days)	Immediate predecessor
A	5	–
B	15	*A*
C	8	*B*
D	9	*C*
E	10	*B*
F	7	*E*
G	9	*F*
H	5	*A*
I	12	*H*
J	16	*G, I*
K	4	*D, J*

a Draw a network for the activities shown in this table.



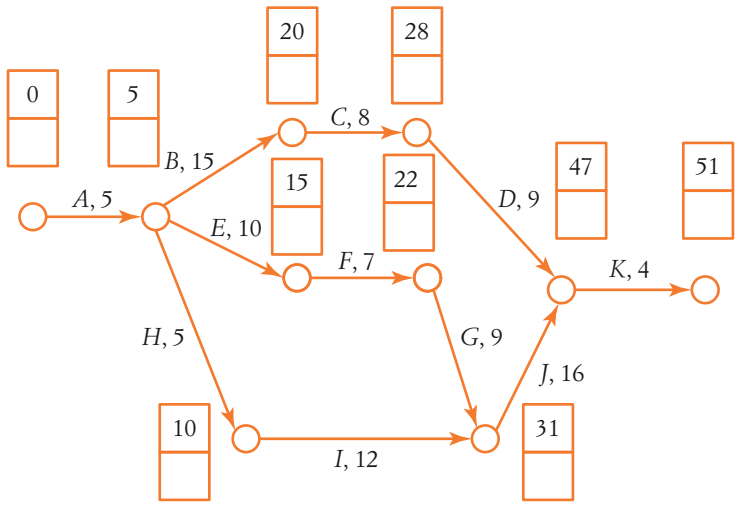
3 marks

b Which activity is the most common immediate predecessor?

B

1 mark

c Complete a forward scan to determine the earliest completion time of the project.

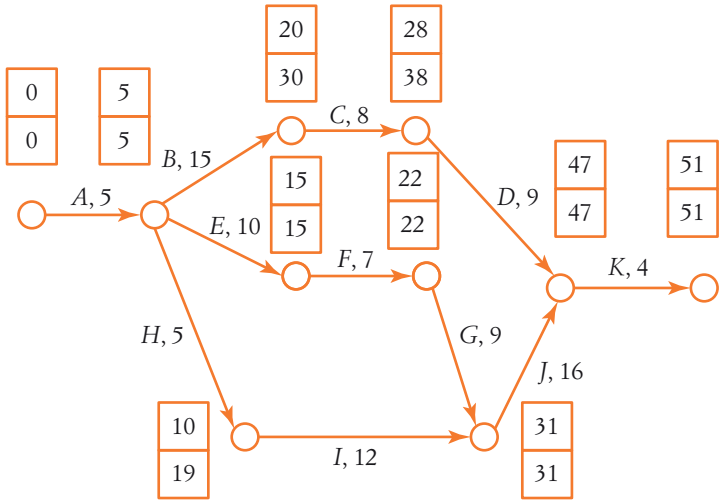


Earliest completion time is 51 days.

3 marks

© Cengage Learning Australia Pty Ltd 2014

d Complete a backward scan and hence find the critical path for the network.



Critical path = *AEFGJK*

3 marks

e List the float times for all non-critical activities and hence determine the activity that can be delayed the longest.

B float time = 30 − 5 − 15 = 10 days

C float time = 38 − 20 − 8 = 10 days

D float time = 47 − 28 − 9 = 10 days

H float time = 19 − 5 − 5 = 9 days

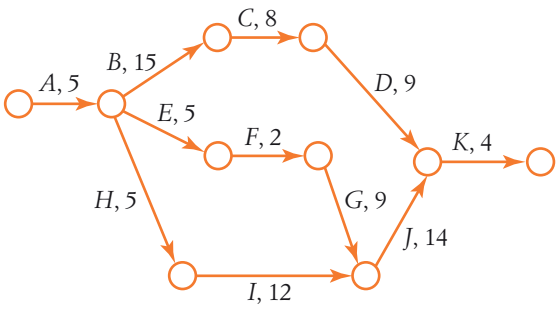
I float time = 31 − 5 − 12 = 14 days

Activity *I* can be delayed the longest.

3 marks

Extra resources are put on so that the time required to complete activities *E* and *F* is reduced by 5 days, and activity *J* is reduced by 2 days.

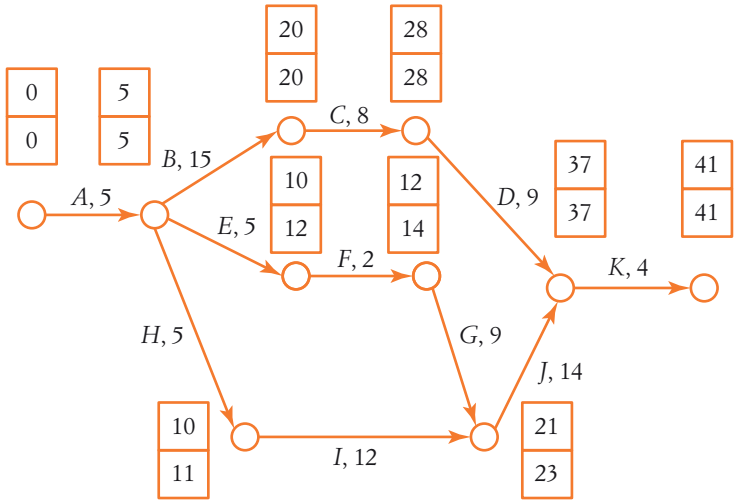
f Redraw the graph to represent these extra resources.



1 mark

© Cengage Learning Australia Pty Ltd 2014

g Determine the new critical path and the new project completion time.



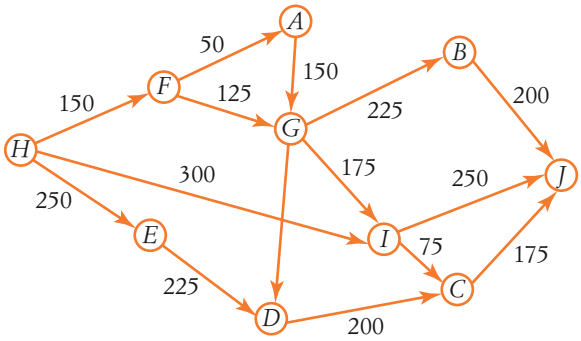
Critical path = *ABCDK*

Project completion time = 41 days.

3 marks

Question 5

Consider this directed network.



a What is the inflow capacity of vertex *C*?

275

1 mark

b What is the outflow capacity of vertex *B*?

100

1 mark

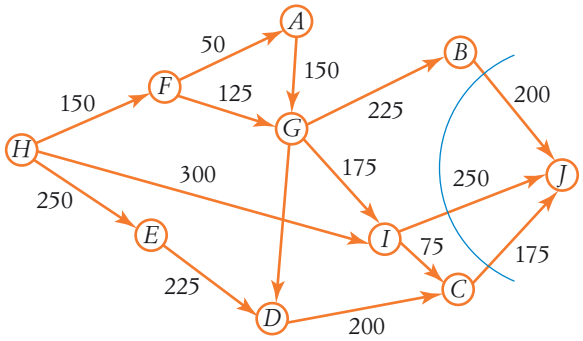
c What is the maximum flow out of vertex *A*?

50

1 mark

© Cengage Learning Australia Pty Ltd 2014

d Determine the maximum flow for this network where source = H and sink = J.



Capacity of cut = 200 + 250 + 175 = 625

Maximum flow = 625

2 marks

Question 6

Consider this table, showing the number of weeks four different companies take to complete four different tasks.

	Task 1	Task 2	Task 3	Task 4
A	13	19	25	16
B	9	13	14	20
C	21	25	27	18
D	22	26	24	15

a Perform a row and column reduction on the table.

Row reduction:

	Task 1	Task 2	Task 3	Task 4
A	0	6	12	3
B	0	4	5	11
C	3	7	9	0
D	7	11	9	0

Column reduction:

	Task 1	Task 2	Task 3	Task 4
A	0	2	7	3
B	0	0	0	11
C	3	3	4	0
D	7	7	4	0

2 marks

© Cengage Learning Australia Pty Ltd 2014

b Is the Hungarian algorithm required? If so, perform this to complete the row and column reductions.

Yes, the zeros can be covered with three lines, which is not equal to the number of rows (4).

	Task 1	Task 2	Task 3	Task 4
A	0	2	7	3
B	0	0	0	11
C	3	3	4	0
D	7	7	4	0

Smallest uncovered number is 3.

	Task 1	Task 2	Task 3	Task 4
A	3	5	10	9
B	3	3	3	17
C	3	3	4	3
D	7	7	4	3

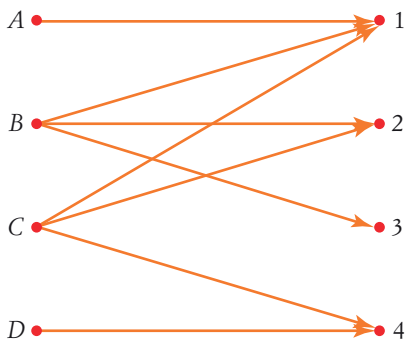
Smallest number is 3.

	Task 1	Task 2	Task 3	Task 4
A	0	2	7	6
B	0	0	0	13
C	0	0	1	0
D	4	4	1	0

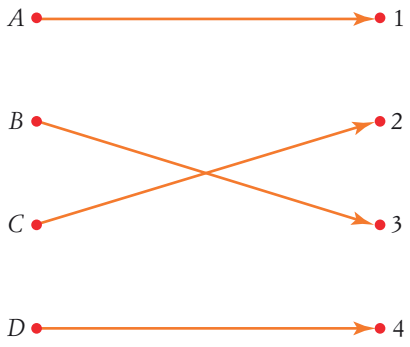
This completes the column and row reduction.

3 marks

c Represent this as a bipartite graph and hence find the optimum allocation.



Optimum allocation:



2 marks

© Cengage Learning Australia Pty Ltd 2014

d Hence, find the minimum:

i total time that it would take to complete all four tasks

Minimum time $= 13 + 14 + 25 + 15 = 67$ weeks.

ii time it would take to complete all four tasks if the four companies are able to work independently of one another.

Minimum time to completion is the time for the longest task $-$ 25 weeks.

2 marks

© Cengage Learning Australia Pty Ltd 2014